MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

END OF SEMESTER EXAMINATION

ICT 4701: SOFTWARE ENGINEERING

TIME ALLOWED: 2 HOURS **DATE : 4th APRIL, 2017**

INSTRUCTIONS

i. This paper contains **Two**pages. Please check.
ii. There are **Two questions** in this paper. Please check.
iii. Answer **ALL** questions.
iv. Marks for each question are indicated.

Question One

a. Discuss**five** components of a design method? **(10 Marks)**

b. What are the **five** benefits of testing software system components individually (unit and module testing) before assembling them into a complete system? **(10 Marks)**

c. Extreme programming express user requirements as stories, with each story written on a card. List and explain **two** advantages and **two** disadvantages of this approach to requirements description. **(4 Marks)**

d. You have been asked to help in the development of a new computerised call centre. Customers from MzuniBankim'khonde will ring up if they wish to change the details of their account. For instance, they might want to open a new account, close an old account, they might want to change the address of the person who owns the account and so on. Briefly explain how you might use **two** different requirements elicitation techniques to identify the potential usability problems that might arise for people working with this new system. Identify the principle strengths and weaknesses of each technique. **(10 Marks)**

e. A company has just designed a user interface that was an utter disaster. You have been brought in as a Software Engineering consultant to explain what they did wrong. For each of the below quotes from the companies engineers, respond and explain what they should have done. What design principle(s) did they violate?

i. "We used it ourselves for over a week! We really loved it! I can't understand why we're getting all these nasty letters from our users!" **(3 Marks)**

ii. "All of us felt that OK and Cancel buttons in the lower right hand corner were boring. So instead, if you click at the top of the dialog box, it closes as if you did "OK," and if you click at the bottom, it closes as if you clicked "Cancel." Labels just take up screen real estate." **(3 Marks)**

iii. "The users seem to be getting stuck on one screen even though the information they need to answer questions was two screens back. Can't they just remember things?" **(4 Marks)**

Question Two

a. What are the **eight** important attributes that all professional software should have. **(8 Marks)**

b. List **two** advantages and **two** disadvantages professional software engineer should be certified in the same way as doctors or lawyers. **(4 Marks)**

c. List any **two** reasons why the productivity rate of programmers working as a pair is roughly the same as two programmers working individually. **(2 Marks)**

d. To help counter terrorism, many countries are planning the development of computer systems that track large numbers of their citizens and their actions. Discuss the ethical implication of developing this type of system. **(2 Marks)**

e. Discuss the following software costing models.
   i. LOC. **(5 Marks)**
   ii. FPA. **(5 Marks)**
   iii. COCOMO I. **(5 Marks)**
   iv. COCOMO II. **(5 Marks)**

END OF QUESTION PAPER